

NADA ABOUBAKR

DATA SCIENTIST | APPLIED AI DEVELOPER

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SUMMARY

Data Scientist and Applied AI Engineer with experience developing ML and LLM-based systems across finance and healthcare. Built secure offline AI assistants, RAG-powered search tools, and a decision-support application that improved data autonomy and operational efficiency. Strong background in data processing, statistical analysis, and dashboard development, with proven contributions to real-world analytics and automation projects.

TECHNICAL SKILLS

Programming: Python, R, C++, Java

ML & AI Tools: TensorFlow, FastAPI, Flask, LlamaIndex, FAISS, Ollama, NER, BLIP, LoRA, RF

Database & Visualization: SQL, SQLite, Oracle, Power BI, MATLAB, PostgreSQL, Data Modeling, Excel

DevOps: Docker, Git, AWS, CI/CD

Languages: Arabic (Native), English (C1)

WORK EXPERIENCE

Applied AI Engineer Intern | Q-Auto, Qatar

Nov 2025 – Present

- Developed a decision-making LLM model for a vehicle financing application, maintained secure connections across the API calls.
- Conducted black-box QA testing for the financing portal and HR system

Data Analyst Institutional Impact Intern | Qatar Foundation, Qatar

Nov 2025 – Present

- Processed and harmonized alumni survey data using Python/Excel, improving data quality and ensuring reliable analysis inputs.
- Applied statistical methods to support the development of alumni personas and collaborated with senior analysts to interpret results and standardize analytical workflows, contributing insights for the WISE Conference.
- Analyzed and reported 2025 QF Alumni Forum feedback to inform improvements for future forums and led the design of the 2025 Alumni Tracer Study for QF alumni.

Applied AI Engineer Intern | Q-Auto, Qatar

Apr 2025 – Jul 2025

- Developed a financial assistant using Google GenAI, automating loan decision support and chatbot Q&A.
- Designed a secure, offline AI assistant with LLM capabilities, reducing dependency on external APIs by 100%.
- Engineered an AI-powered internal search engine using custom embeddings, improving internal document retrieval.
- Integrated RAG architecture using LlamaIndex, FAISS, and Ollama to deliver high accuracy chatbot responses.
- Implemented a conversation memory database for the AI assistant, evaluating SQLite, MongoDB, and PostgreSQL for scalability and efficiency.

IBM Data Scientist Trainee | Digital Egypt Pioneers Initiative (DEPI), Egypt

Apr 2024 – Oct 2024

- Completed a 6-month data science program focusing on machine learning and IBM technologies, with training in Python, SQL, model building, data pipelines, and visualization.
- Developed a machine learning model to predict the ten-year risk of coronary heart disease using the Framingham Heart Study dataset, achieving 89% accuracy, with a GUI for simplified user interaction. | [GitHub](#)

IT & Telecom (Data Analysis) Intern | Al-Ahli Hospital, Qatar

Aug 2023 – Sep 2023

- Delivered 3 Power BI dashboards that improved strategic decision-making for hospital operations.
- Applied basic ML models to predict demographic health trends, increasing data utilization efficiency.

Machine Learning Intern | Technocolabs Softwares Inc., Remote

Apr 2023 – Jul 2023

- Increased mortgage trend prediction accuracy by 86% using supervised ML and deployed with Flask.
- Engineered a retail sales forecasting model (70% accuracy) by leading feature design and exploratory analysis informing SKU-level inventory decisions.

EDUCATION

Master of Data Analytics in Health Management, Hamad Bin Khalifa University, Qatar

2025 - 2027

Bachelor of Computer and Data Science, Alexandria University, Egypt – CGPA: 3.407/4

2020 - 2024

SELECTED PROJECTS

Drug Combination Safety Predictor

[GitHub](#)

- Built a deep learning model with AUPR of 0.915 to forecast drug-drug interactions from gene expression data.
- Constructed 978-dimensional vectors using gene properties, increasing interpretability and clinical value.
- Deployed system on cloud with real-time web interface; used by medical researchers.

Suicidal Post Detection (Arabic NLP)

- Designed an NLP model to flag suicidal ideation in Modern Standard Arabic posts, aiding mental health awareness.

CERTIFICATION

IELTS Academic: Band Score 7.5

Dec 2024

DataCamp Associate Data Analyst in SQL

Oct 2024

ADDITIONAL EXPERIENCE

Wavemakers United – Web Developer Volunteer

- Designed and developed user-friendly websites focused on improving usability and accessibility for clients.

IEEE SSCS Alex SC – Machine Learning & HR Volunteer

- Presented an internal session on data visualization.
- Managed events and led volunteer recruitment and onboarding.

Weill Cornell Medicine – Qatar – Volunteer for the AI Center for Precision Health

- Analyzed accelerometer data and built predictive models based on it.
- Collaborated with the team to investigate the RADAR system and help get it running on the shared computing system.